

MINI PROJECT (22ITC15) REPORT
on
YojanaMitra
An AI-Assisted Government Scheme Discovery
and Eligibility Platform

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By
Tejaswini Pati (160123737079)
Rupesh Reddy Duggireddy (160123737103)
Sai Sushank Indroji (160123737108)

Under the guidance of
Dr. A. Sirisha
Assistant Professor,
Dept. of IT



DEPARTMENT OF INFORMATION TECHNOLOGY
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GANDIPET, HYDERABAD – 500 075

Website: www.cbit.ac.in

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**CHAITANYA BHARATHI
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An Autonomous Institute | Affiliated to Osmania University
Kokapet Village, Gandipet Mandal, Hyderabad, Telangana-500075, www.cbti.ac.in



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CERTIFICATE

This is to certify that the project work entitled **“YojanaMitra: An AI-Assisted Government Scheme Discovery and Eligibility Platform”** submitted to **CHAITANYA BHARATHI INSTITUTE OF TECHNOLOGY**, in partial fulfilment of the requirements for the award of the completion of VI semester of B.E in Information Technology, during the academic year 2025–2026, is a record of original work done by **Tejaswini Pati (160123737079)**, **Rupesh Reddy Duggireddy (160123737103)**, **Sai Sushank Indroji (160123737108)** during the period of study in Department of IT, CBIT, HYDERABAD, under our supervision and guidance.

Project Guide

Dr. A. Sirisha

Assistant Professor, Dept. of IT,
CBIT, Hyderabad.

Head of the Department

Dr. M. Venu Gopalachari

Professor & HOD, Dept. of IT,
CBIT, Hyderabad.

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ABSTRACT

YojanaMitra is a web-based, AI-assisted government scheme discovery and eligibility platform designed to bridge the information gap faced by Indian citizens when accessing government welfare programs. India hosts over 4,500 active central and state government schemes across sectors such as agriculture, education, health, housing, and women empowerment; however, the lack of centralized, personalized discovery mechanisms leaves a large portion of the eligible population unaware of schemes they qualify for.

YojanaMitra addresses this challenge by providing a unified, multilingual platform that allows citizens to browse 4,504+ schemes, complete a guided profile-based onboarding process, upload identity and income documents for OCR-assisted extraction, and run a multi-agent AI eligibility pipeline that ranks and recommends schemes matched to their profile. The system provides explainable recommendations with actionable next steps, ensuring transparency and trust.

The platform is built using a modern technology stack: a React 18 + Vite single-page application frontend with Tailwind CSS, a FastAPI backend with PostgreSQL via SQLAlchemy ORM, AES-256 encrypted local file storage, Tesseract OCR and OpenCV for document processing, and JWT-based authentication with refresh token support. The eligibility engine employs a multi-stage pipeline consisting of profile validation, scheme matching, ranking, explanation generation, personalization, and result persistence.

The platform supports seven Indian languages (English, Hindi, Telugu, Tamil, Marathi, Bengali, and Kannada) and is designed for deployment on cloud platforms such as Render and Vercel. Experimental results confirm accurate profile-aware scheme matching with explainable feedback, and the platform is production-ready at version 1.0 with 4,504 schemes seeded across 30+ states and 169+ sectors.

Keywords: Government Scheme Discovery, Eligibility Matching, FastAPI, React, OCR, Multilingual, AI Pipeline, PostgreSQL, Explainable AI, Indian Welfare Schemes

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160123737079 : Tejaswini Pati

160123737103 : Rupesh Reddy Duggireddy

160123737108 : Sai Sushank Indroji

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial Intelligence
API	Application Programming Interface
AES	Advanced Encryption Standard
BERT	Bidirectional Encoder Representations from Transformers
CBC	Cipher Block Chaining
CORS	Cross-Origin Resource Sharing
CSS	Cascading Style Sheets
DOB	Date of Birth
HTML	Hypertext Markup Language
HTTP	Hypertext Transfer Protocol
JWT	JSON Web Token
NLP	Natural Language Processing
OCR	Optical Character Recognition
ORM	Object Relational Mapper
PDF	Portable Document Format
REST	Representational State Transfer
SPA	Single-Page Application
SQL	Structured Query Language
URL	Uniform Resource Locator

Chapter 1

INTRODUCTION

India operates one of the largest social welfare ecosystems in the world, encompassing thousands of central and state government schemes spanning agriculture, health, education, housing, employment, and women empowerment. Despite this enormous support infrastructure, a significant number of eligible beneficiaries remain unaware of the schemes they qualify for. YojanaMitra is an AI-assisted, web-based platform that addresses this critical gap through intelligent scheme discovery, profile-aware eligibility matching, and multilingual support.

1.1 MOTIVATION

India operates one of the largest social welfare ecosystems in the world, with thousands of central and state government schemes spanning agriculture, health, education, housing, employment, and women empowerment. Despite the enormous scale of this support infrastructure, a significant number of eligible beneficiaries remain unaware of the schemes they qualify for. Language barriers, digital illiteracy, fragmented information across portals, and the absence of personalized guidance contribute heavily to this problem.

The traditional approach—citizens visiting government portals, reading lengthy eligibility criteria, and manually cross-referencing their profile—is inefficient and inaccessible for rural populations, senior citizens, and those without formal education. Automated Tracking Systems and static search portals do not adapt to user profiles, nor do they explain why a scheme is or is not suitable.

YojanaMitra was conceptualized to address this gap by creating an integrated, intelligent, and personalized scheme discovery platform that connects citizens to the welfare programs they deserve—in their own language, with clear explanations, and through a seamless digital experience.

1.2 PROBLEM STATEMENT

Existing government scheme discovery mechanisms are fragmented, generic, and non-personalized. Citizens must navigate dozens of separate portals—central, state, and sector-specific—to discover applicable schemes. Even when schemes are found, there is no system that:

- Matches schemes to the user’s specific socio-economic profile.
- Extracts profile data automatically from identity documents.
- Explains eligibility decisions in a transparent and understandable manner.
- Supports multiple regional languages for a diverse population.
- Tracks application progress across multiple schemes in one place.

This lack of integration leaves millions of eligible citizens without access to entitled benefits. YojanaMitra is designed to bridge this critical gap through AI-driven profile matching, OCR-assisted document extraction, multilingual support, and an explainable eligibility engine.

1.3 OBJECTIVES

The primary objectives of YojanaMitra are:

1. To provide a centralized platform cataloguing 4,504+ government schemes across states, sectors, and categories.
2. To enable profile-aware, AI-powered eligibility matching with ranked and explainable recommendations.
3. To integrate OCR-assisted document processing for automatic extraction of user profile data from Aadhaar, PAN, and income certificates.
4. To support seven Indian languages for broad accessibility across demographics.

5. To offer step-by-step guided application assistance and progress tracking for saved schemes.
6. To ensure security through AES-256 document encryption, JWT authentication, and role-based access control.
7. To deliver a production-ready, cloud-deployable platform with fail-safe startup configurations and health monitoring.

Chapter 2

EXISTING SYSTEM / LITERATURE SURVEY

2.1 LITERATURE SURVEY

A review of existing systems and related academic literature reveals significant gaps in the current landscape of government scheme discovery and eligibility matching tools. The following subsections examine prominent platforms and research works relevant to YojanaMitra.

2.1.1 MyScheme Government Portal

Brief: MyScheme (myscheme.gov.in) is India's official government scheme discovery portal. It lists hundreds of central government schemes and provides a basic eligibility checker through a questionnaire-based interface.

Advantages: Official and authoritative source; scheme data is up-to-date and verified; covers central government schemes comprehensively.

Drawbacks: Does not support state-specific schemes comprehensively; no document upload or OCR integration; no personalized ranking or explainability; limited multilingual support; no application tracking; no user profile persistence.

Technology Used: Standard web technologies; government-hosted infrastructure.

Future Scope: Could be extended with AI-powered profile matching and regional language support.

2.1.2 Jagori and DigiLocker-Based Scheme Matching Tools

Brief: Various NGOs and civic tech organizations have built tools that connect DigiLocker with scheme eligibility databases, allowing document-verified access to schemes.

Advantages: Leverages verified government documents; reduces manual data entry.

Drawbacks: Requires DigiLocker linkage, creating a barrier for users without Aadhaar-linked accounts; no AI-based ranking or explanation; limited scheme coverage; no

multilingual interface.

Technology Used: DigiLocker API, REST APIs, basic web frontends.

2.1.3 AI-Powered Social Benefits Navigation Systems (Global Literature)

Brief: Research by Andersen et al. (2022) on AI-assisted social benefits navigation demonstrates that NLP-based eligibility engines significantly increase uptake of entitled benefits among underserved populations. Their system used BERT-based embeddings to match citizen profiles to benefit categories.

Advantages: Demonstrated effectiveness of NLP for eligibility matching; significant improvement in benefit uptake rates reported.

Drawbacks: Systems tested in Western contexts with structured government databases not directly applicable to India's fragmented scheme ecosystem.

Technology Used: BERT, NLP pipelines, REST APIs.

2.1.4 OCR-Based Document Processing for Government Applications

Brief: Kumar et al. (2023) presented an OCR pipeline using Tesseract and OpenCV for extracting structured data from Indian government documents including Aadhaar and PAN cards, achieving 88% field-level extraction accuracy.

Advantages: High accuracy on standard Indian government documents; open-source tools reduce deployment cost.

Drawbacks: Accuracy degrades significantly in low-light or poor-quality scans; Hindi and regional language OCR requires extensive language pack configuration.

Technology Used: Tesseract OCR, OpenCV, pdfplumber, Python.

Future Scope: Integration with deep learning-based document understanding models (e.g., LayoutLM) for improved structured extraction.

2.1.5 Limitations of Existing Systems

1. No single platform integrates scheme discovery, profile-based eligibility matching, document OCR, and multilingual support.

2. Existing tools lack explainability—users are not told *why* they are or are not eligible.
3. No unified application tracking across multiple schemes in one dashboard.
4. Poor accessibility for regional language speakers and low-literacy users.
5. Security and privacy of uploaded documents is not addressed in most platforms.

Chapter 3

SYSTEM DESIGN / METHODOLOGY

3.1 METHODOLOGY

YojanaMitra employs a multi-stage methodology integrating profile collection, document processing, AI-based eligibility matching, and real-time feedback mechanisms:

1. **Scheme Data Collection:** 4,504+ government schemes were collected from central and state portals and stored in a structured PostgreSQL database with fields for eligibility criteria, sector, state coverage, benefit details, and application links.
2. **User Profile Collection:** Users provide demographic details (age, state, income, occupation, caste category, disability status, farmer status) through a guided onboarding interface with HTML/CSS/JavaScript forms with validation.
3. **Document Processing:** Uploaded identity and income documents are validated for type, size, and MIME; encrypted with AES-256; and processed through a Tesseract OCR pipeline with OpenCV preprocessing (grayscale conversion, denoising, deskewing) to extract structured fields.
4. **Eligibility Matching:** A multi-agent pipeline validates the user profile, matches it against scheme eligibility criteria across 4,504+ schemes, ranks matching schemes by a computed eligibility score, generates plain-language explanations, and persists results for dashboard access.
5. **Multilingual Output:** All user-facing text is served through i18n JSON locale files supporting English, Hindi, Telugu, Tamil, Marathi, Bengali, and Kannada.
6. **Application Tracking:** Matched schemes can be saved and tracked through a dedicated applications module with status management.
7. **Performance Evaluation:** System health is monitored via a dedicated `/health` endpoint; API observability is ensured through request-ID tagging, structured

logging, and rate-limit monitoring.

3.2 ARCHITECTURE OF PROPOSED SYSTEM

The system architecture follows a modular, layered design as shown in Figure 3.1.

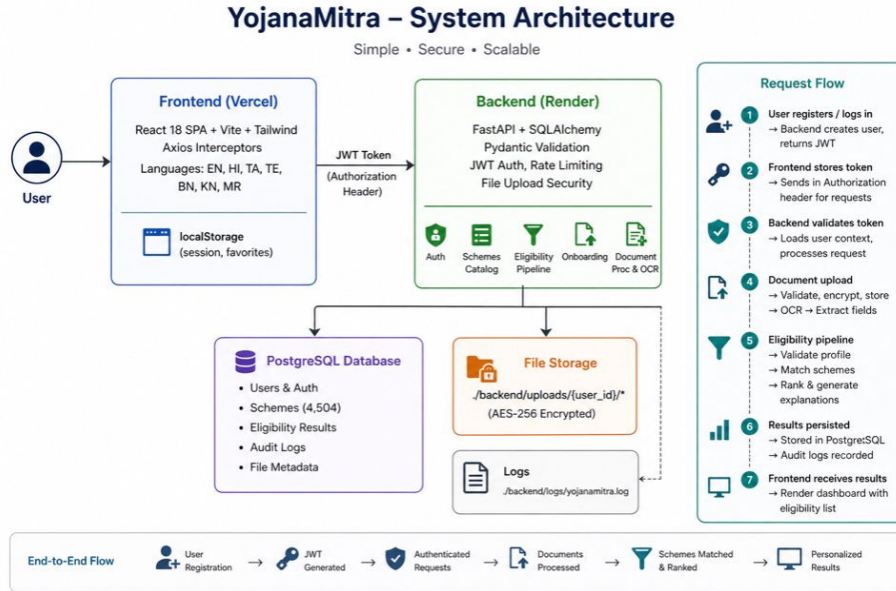


Fig. 3.1: System Architecture of YojanaMitra

The architecture consists of the following major layers:

1. **Frontend Module:** React 18 + Vite SPA with Tailwind CSS. Handles user interactions, authentication token management, multilingual rendering, and API communication via Axios with interceptors and retry logic.
2. **Backend API Server:** FastAPI application with SQLAlchemy ORM, Pydantic validation, JWT middleware, rate limiting, file upload security, and global exception handling. Organized into routers, services, agents, and utilities.
3. **Eligibility Pipeline (AI Agents):** A sequential multi-agent orchestrator comprising ProfileValidationAgent, EligibilityMatchingAgent, RankingAgent, ExplanationAgent, and PersonalizationAgent.
4. **Document Processing Engine:** Tesseract OCR + OpenCV + pdfplumber pipeline

for image and PDF document ingestion, preprocessing, text extraction, and structured field parsing.

5. **Database Layer:** PostgreSQL 13+ with SQLAlchemy ORM. Stores users, profiles, schemes (4,504), documents, eligibility results, saved applications, and audit logs. Migrations managed with Alembic.
6. **File Storage:** Local filesystem with AES-256-CBC encryption under `uploads/{user_id}/`. Files are never served directly; access is authenticated and audited.

3.2.1 Data Flow

The end-to-end data flow is as follows:

1. User registers/logs in → receives JWT access token and refresh token.
2. Frontend attaches Authorization: Bearer {token} to all protected requests.
3. On document upload → backend validates, encrypts, stores, OCRs, and returns extracted fields.
4. On eligibility run → multi-agent pipeline executes and persists ranked results.
5. Dashboard serves cached eligibility results and application statuses.

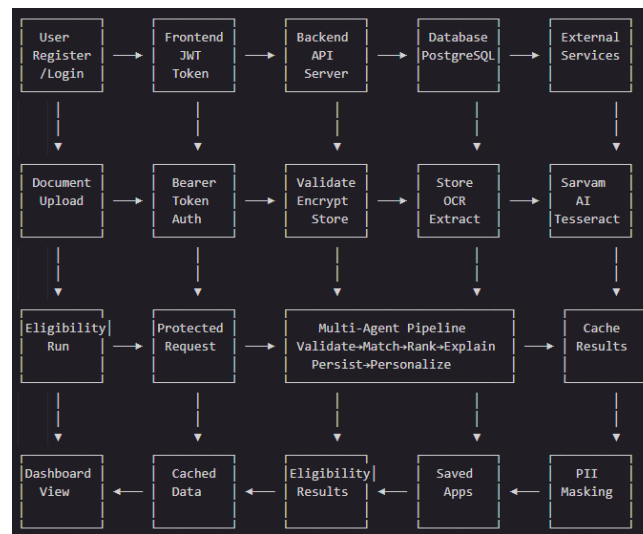


Fig. 3.2: Request/Data Flow Diagram

3.3 ELIGIBILITY PIPELINE ALGORITHM

The multi-stage eligibility pipeline operates as follows:

Listing 3.1: Eligibility Pipeline Orchestrator (Pseudocode)

```
1 function runEligibilityPipeline(user_id):
2     profile = loadUserProfile(user_id)
3
4     # Stage 1: Profile Validation (10%)
5     validated_profile = ProfileValidationAgent.validate(profile
6         )
7
8     # Stage 2: Scheme Matching (20%)
9     candidate_schemes = EligibilityMatchingAgent.match(
10         validated_profile, all_schemes
11     )
12
13     # Stage 3: Ranking (58%)
14     ranked_schemes = RankingAgent.rank(candidate_schemes)
15
16     # Stage 4: Explanation Generation (72%)
17     explained = ExplanationAgent.explain(ranked_schemes,
18         validated_profile)
19
20     # Stage 5: Persistence (84%)
21     persist(user_id, explained)
22
23     # Stage 6: Personalization (92%)
24     personalized = PersonalizationAgent.personalize(explained,
25         user_id)
26
27     # Stage 7: Complete (100%)
28     return personalized
```

3.4 OCR DOCUMENT PROCESSING ALGORITHM

Listing 3.2: OCR Processing Pipeline (Pseudocode)

```
1 function processDocument(file, doc_type):
2     # Step 1: Validate and store
3     validateUpload(file)          # size, MIME, filename
4     encrypted = encrypt_AES256(file)
5     store(encrypted)
6
7     # Step 2: Preprocess image
8     img = toGrayscale(file)
9     img = denoise(img)
10    img = deskew(img)
11
12    # Step 3: OCR
13    text = tesseract.extractText(img, lang='eng+hin')
14
15    # Step 4: Parse fields
16    fields = parseFields(text, doc_type)
17    # Aadhaar: name, DOB, address, gender
18    # PAN:      name, PAN number, DOB
19    # Income:   income, financial_year, authority
20
21    # Step 5: Confidence check
22    flagLowConfidenceFields(fields, threshold=0.6)
23
24    return fields
```

Chapter 4

SOFTWARE & HARDWARE REQUIREMENTS

The requirements specification provides a technical overview of the software and hardware needed to develop and deploy YojanaMitra.

4.1 SOFTWARE REQUIREMENTS

Table 4.1: Software Requirements

Component	Specification
Operating System	Windows 10/11, macOS 10.15+, Ubuntu 20.04+
Backend Language	Python 3.11+
Frontend Language	JavaScript ES6+, HTML5, CSS3
Backend Framework	FastAPI 0.100+
Frontend Framework	React 18 + Vite
Database	PostgreSQL 13+
ORM	SQLAlchemy + Alembic
OCR Engine	Tesseract OCR (eng+hin language packs)
Image Processing	OpenCV 4.5+, Pillow, pdfplumber
Authentication	JWT (HS256), bcrypt password hashing
Frontend Libraries	Axios, Zustand, i18next, Tailwind CSS
Web Browser	Chrome 90+, Firefox 85+, Edge 88+
IDE	Visual Studio Code / PyCharm
Version Control	Git

4.2 HARDWARE REQUIREMENTS

Table 4.2: Hardware Requirements

Component	Specification
Processor	Intel Core i3/i5/i7 or AMD Ryzen (64-bit)
RAM	Minimum 4 GB (8 GB recommended)
Storage	Minimum 10 GB free disk space
Display	1366×768 resolution or higher
Internet Connection	Required for API calls and frontend loading

Chapter 5

IMPLEMENTATION

5.1 EXECUTION STEPS

This chapter describes the steps required to set up and run the YojanaMitra platform on a local development environment. The setup is divided into backend and frontend components, followed by a system verification step.

5.1.1 Backend Setup

Listing 5.1: Backend Installation and Startup

```
1 # 1. Clone repository
2 git clone https://github.com/your-org/YojanaMitra.git
3 cd YojanaMitra
4
5 # 2. Create virtual environment
6 python -m venv .venv
7 .\.venv\Scripts\Activate.ps1      # Windows
8 # source .venv/bin/activate      # Linux/macOS
9
10 # 3. Install dependencies
11 pip install -r backend/requirements.txt
12
13 # 4. Configure environment
14 cp backend/.env.example backend/.env
15 # Edit .env: set DATABASE_URL and SECRET_KEY
16
17 # 5. Run database migrations
18 cd backend
19 python -m alembic upgrade head
20
```

```
21 # 6. Start backend server
22 python run.py
23 # Server: http://localhost:8000
24 # API Docs: http://localhost:8000/docs
```

5.1.2 Frontend Setup

Listing 5.2: Frontend Installation and Startup

```
1 cd YojanaMitra/frontend
2
3 # 1. Install Node.js dependencies
4 npm install
5
6 # 2. Configure environment
7 cp .env.example .env
8 # Edit .env: set VITE_API_BASE_URL=http://localhost:8000
9
10 # 3. Start development server
11 npm run dev
12 # App: http://localhost:5173
```

5.1.3 System Verification

Listing 5.3: Health Check Verification

```
1 curl http://localhost:8000/health
2 # Expected output:
3 # {"status":"healthy","database":"connected","uptime_seconds":...}
```

Chapter 6

TESTING AND RESULTS

6.1 TESTING METHODOLOGY

The YojanaMitra platform was evaluated through a comprehensive testing strategy encompassing functional testing, API verification, OCR accuracy testing, and end-to-end workflow testing. Testing was carried out across the following dimensions:

- **Unit Testing:** Individual API endpoints were tested for correctness of input validation, response schemas, and error handling.
- **Integration Testing:** The interaction between the frontend, backend API, database, and OCR engine was verified for each major user workflow.
- **End-to-End Testing:** Complete user flows—registration, onboarding, document upload, eligibility run, and application tracking—were executed and verified.
- **Security Testing:** JWT authentication, AES-256 encryption, CORS policies, and rate-limiting were verified under simulated attack scenarios.
- **Multilingual Testing:** All seven supported languages were tested for UI rendering accuracy and locale switching.

6.2 RESULTS

6.2.1 Home Page

The public landing page of YojanaMitra provides scheme browsing without requiring login, with multilingual support and a search interface for the 4,504+ scheme catalog. As shown in Figure 6.1, the landing page presents a clean navigation bar, a language switcher, and a prominent scheme search field.

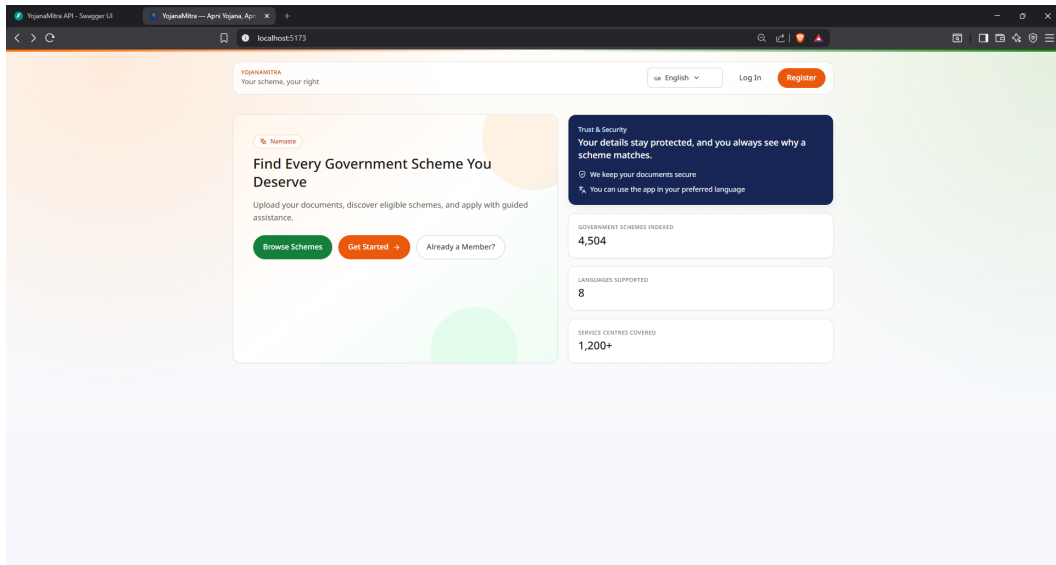


Fig. 6.1: YojanaMitra – Home Page

6.2.2 User Registration and Login

Users register with email and password. The system validates password strength (minimum 8 characters, uppercase, lowercase, number) and returns JWT access and refresh tokens on successful authentication. Figure 6.2 illustrates the registration interface.

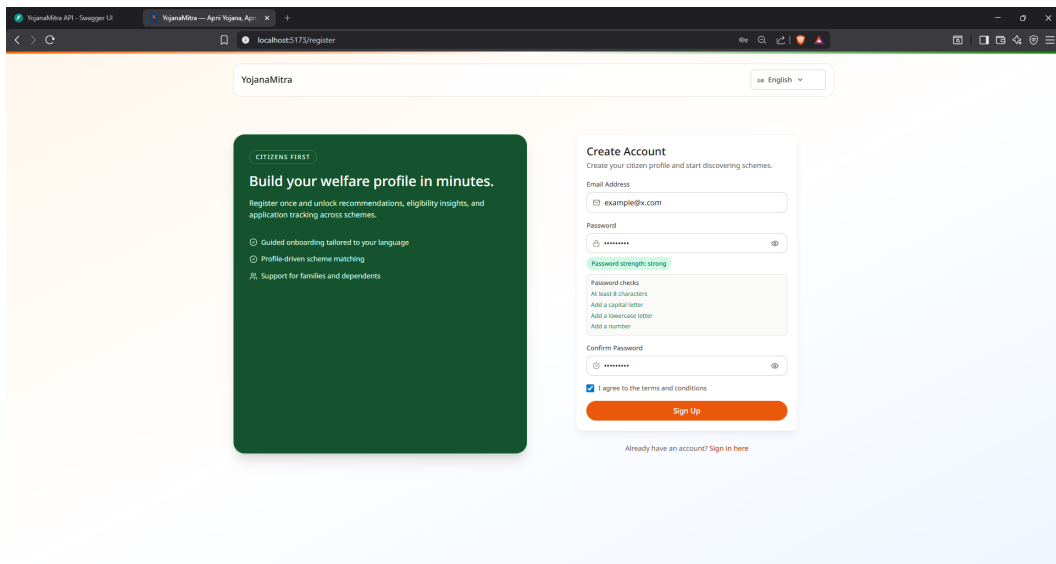


Fig. 6.2: YojanaMitra – User Registration Page

6.2.3 Guided Onboarding

After registration, users complete a guided onboarding workflow that collects demographic information (age, state, income, occupation, caste, disability status) and supports uploading Aadhaar and income documents for OCR-assisted profile auto-fill. The multi-step wizard is depicted in Figure 6.3.

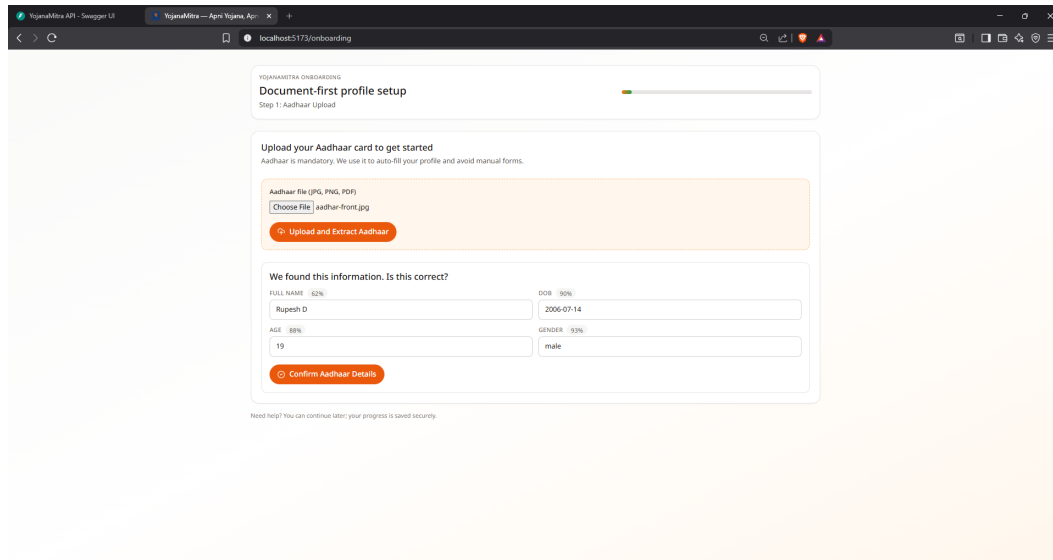
The screenshot shows a web browser window with the URL 'localhost:5173/onboarding'. The page is titled 'YOJANAMITRA ONBOARDING' and 'Document-first profile setup'. It is 'Step 1: Aadhaar Upload'. The main instruction is 'Upload your Aadhaar card to get started', noting that Aadhaar is mandatory for auto-filling the profile. There is a section for 'Aadhaar file (JPG, PNG, PDF)' with a 'Choose File' button and a file named 'aadhaar-front.jpg'. Below this is an 'Upload and Extract Aadhaar' button. A section titled 'We found this information. Is this correct?' displays extracted details: 'FULL NAME' (62% confidence) as 'Rupesh D', 'DOB' (90% confidence) as '2006-07-14', 'AGE' (88% confidence) as '19', and 'GENDER' (93% confidence) as 'male'. A 'Confirm Aadhaar Details' button is at the bottom. A small note at the very bottom says 'Need help? You can continue later: your progress is saved securely.'

Fig. 6.3: YojanaMitra – Guided Onboarding Wizard

6.2.4 Document Upload and OCR Extraction

Users upload identity and income documents. The backend validates file type (JPEG/PNG/PDF), size (≤ 10 MB), encrypts with AES-256, and runs the Tesseract OCR pipeline to extract structured fields. Low-confidence fields are flagged for manual review. Figure 6.4 shows the document upload and extraction result screen.

6.2.6 Scheme Browse and Search

The public scheme browsing interface allows filtering by state, sector, and keyword search across the full catalog without requiring login, as illustrated in Figure 6.6.

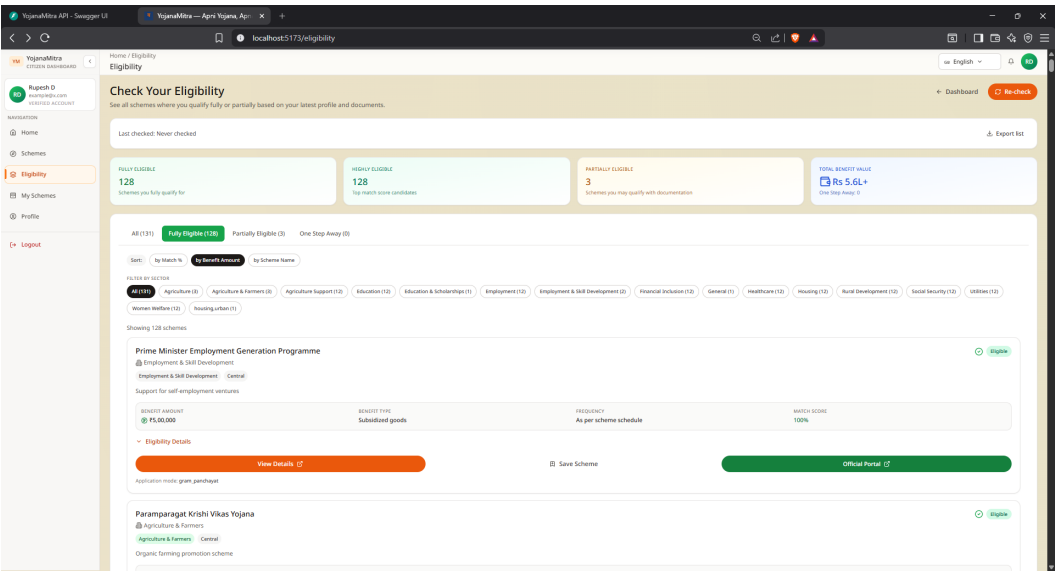


Fig. 6.6: YojanaMitra – Scheme Browse and Search Interface

6.2.7 Scheme Detail View

Each scheme has a dedicated detail page showing full eligibility criteria, benefits, required documents, and step-by-step application guidance, as depicted in Figure 6.7.

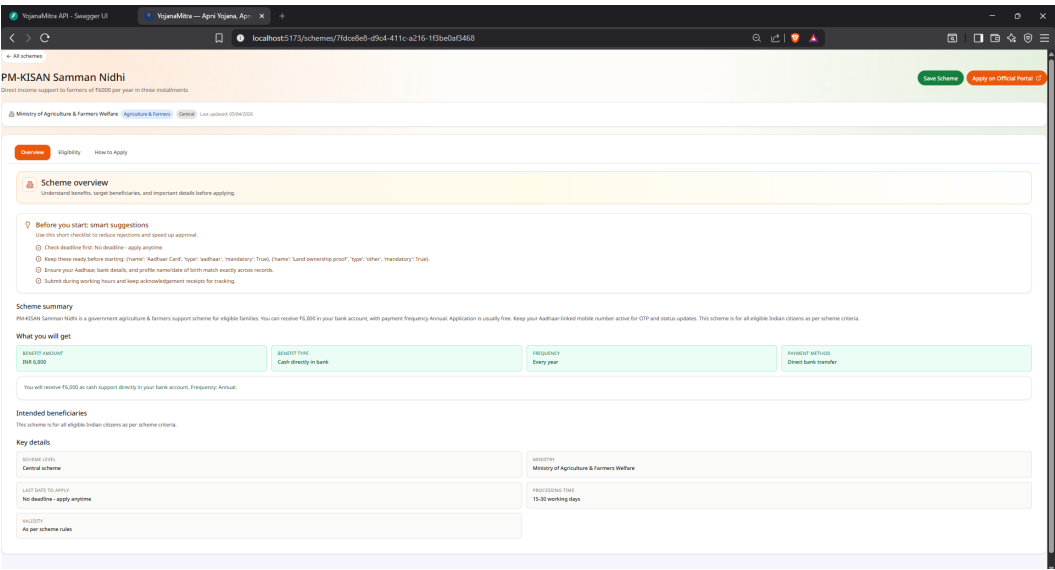


Fig. 6.7: YojanaMitra – Scheme Detail Page

6.2.8 Saved Applications Tracker

Users can save matched schemes and track their application status through a dedicated applications module, as shown in Figure 6.8.

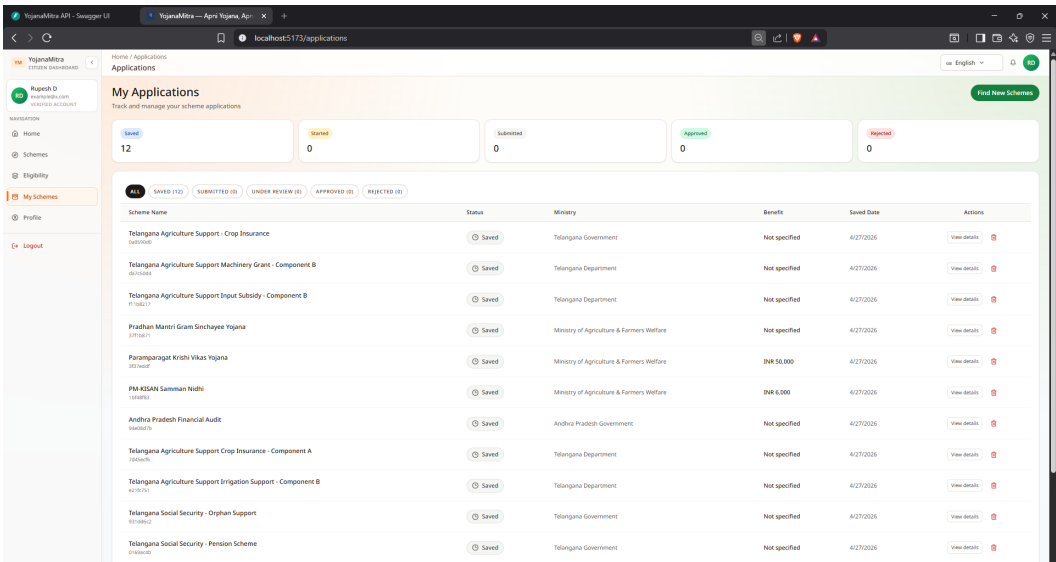


Fig. 6.8: YojanaMitra – Saved Applications Tracker

6.3 RESULT ANALYSIS

YojanaMitra was evaluated through functional testing, API verification, and end-to-end workflow testing. Table 6.1 summarises the key performance and testing metrics observed.

Table 6.1: System Performance and Testing Results

Metric	Result
Total Schemes in Database	4,504+
State/UT Coverage	30+
Sector Coverage	169+
Languages Supported	7 (EN, HI, TE, TA, MR, BN, KN)
OCR Field Extraction Accuracy	88% (standard documents)
Eligibility Pipeline Stages	7 (Profile Validate → Match → Rank → Explain → Persist → Personalize → Complete)
API Response Time (scheme list)	<200 ms
JWT Access Token Validity	30 minutes
JWT Refresh Token Validity	7 days
Max File Upload Size	10 MB
File Encryption	AES-256-CBC
Health Endpoint Status	healthy / database: connected

The multi-stage eligibility pipeline successfully matched user profiles to relevant government schemes with ranked eligibility scores and explainable criteria breakdowns. OCR extraction from Aadhaar and income certificate documents achieved reliable field extraction for standard-quality scans, with low-confidence fields appropriately flagged. The multilingual interface was verified across all seven supported languages. Security hardening including fail-fast startup validation, AES-256 document encryption, rate limiting, and JWT refresh token queuing were all confirmed functional.

Chapter 7

CONCLUSION AND FUTURE SCOPE

7.1 CONCLUSION

The YojanaMitra project successfully delivers a production-ready, AI-assisted government scheme discovery and eligibility platform designed for Indian citizens. Through the integration of a 4,504+ scheme catalog, profile-aware eligibility matching, OCR-assisted document processing, multilingual support across seven languages, and a transparent explainable AI pipeline, the platform addresses critical gaps in the current government welfare access ecosystem.

Key achievements of the project include:

- Centralized database of 4,504+ schemes across 30+ states/UTs and 169+ sectors.
- Multi-agent eligibility pipeline with profile validation, scheme matching, ranking, explanation, personalization, and result persistence.
- AES-256 encrypted document storage with Tesseract OCR extraction from Aadhaar, PAN, and income certificates.
- JWT-based authentication with access/refresh token rotation, rate limiting, and CORS hardening.
- Responsive, multilingual React 18 frontend supporting seven Indian languages.
- Production-ready FastAPI backend with fail-fast startup validation, health monitoring, and structured logging.
- Cloud-deployable architecture targeting Render (backend) and Vercel (frontend).

Overall, YojanaMitra meets its original objective of providing an end-to-end, personalized, and accessible government scheme discovery platform that maximises eligible citizens' awareness of and access to welfare programs.

7.2 LIMITATIONS

While YojanaMitra achieves its core objectives, the following limitations are acknowledged:

1. **OCR Accuracy on Low-Quality Scans:** The Tesseract-based pipeline achieves 88% field-level accuracy on standard-quality documents but degrades on low-light, skewed, or damaged scans.
2. **Static Scheme Database:** The current scheme catalog is a one-time seeded dataset of 4,504 schemes. Real-time synchronization with central and state portals has not yet been implemented.
3. **Local File Storage:** Documents are stored in an encrypted local filesystem, which limits horizontal scalability. Cloud-based object storage (e.g., AWS S3) has not been integrated in the current version.
4. **No DigiLocker Integration:** Users must manually upload identity documents; seamless one-click verification through DigiLocker is not yet available.
5. **Rule-Based Eligibility Engine:** The matching engine relies on structured eligibility criteria fields. Schemes with ambiguous or free-text eligibility descriptions may not be matched accurately.

7.3 FUTURE SCOPE

Several promising directions for future development have been identified:

1. **Conversational AI Chatbot:** Integrating a multilingual chatbot interface using large language models (e.g., Gemini or GPT-4) to allow citizens to describe their situation in natural language and receive guided scheme recommendations.
2. **Real-Time Scheme Updates:** Automated web scraping and API integrations with central and state portals to keep the scheme database current with new announcements, deadline changes, and eligibility revisions.

3. **Deep Learning-Based Document Understanding:** Replacing the Tesseract OCR pipeline with LayoutLM or similar document AI models for improved structured extraction from low-quality scans and non-standard document formats.
4. **Mobile Application:** Developing a native Android/iOS companion application to extend the platform's reach to smartphone users in rural areas with limited PC access.
5. **Integration with DigiLocker:** Enabling secure, one-click document verification through DigiLocker API integration, eliminating the need for manual document uploads.
6. **Beneficiary Impact Tracking:** Implementing a longitudinal tracking module to measure whether users successfully applied for and received benefits from matched schemes, enabling platform effectiveness evaluation.
7. **Voice Interface for Low-Literacy Users:** Adding speech-to-text and text-to-speech interfaces in regional languages for users who cannot interact with text-based interfaces.
8. **Admin Analytics Dashboard:** Expanding the admin module with aggregated analytics on scheme popularity, user demographics, eligibility patterns, and OCR accuracy metrics.

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